

Describing Is the New Doing

Building your own software with Claude Code

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A guest session with Dr. Elise Lambert's 2026 cohort

Who I am (30 seconds)

- **VP, Data Solutions AI at FactSet** — I build AI teams that ship real software
- **Visiting Professor, University of South Alabama** — I teach PhD students how to use ML and Python
- **PhD in Security Studies** (University of Central Florida); ML/NLP researcher before industry
- **Research domains I publish in** — political conflict, civil-military relations, NLP & conflict forecasting, applied economics, political psychology

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A question

Has there ever been a report, a dashboard, or a tool you wished existed — and either you lived without it, or you waited weeks for IT to build it?

That is the gap

For decades, if you wanted custom software, you had to:

- **Hire a developer** — expensive, slow, hard to direct
- **Wait for IT** — slow, often deprioritized
- **Live without it** — most common outcome

Result: thousands of small, useful tools never got built. Decisions made with worse information.

Paradigm shift

Old way — you tell a developer what you want. They go away. They come back later. You iterate slowly.

New way — you tell the AI what you want. It builds it **while you watch**. You see it work. You ask for changes. It changes it.

It's like having a **junior analyst sitting next to you** who can read, write, and run code — and never gets tired, never goes to lunch, never costs \$90,000 a year.

What it can do (plain language)

- **Read** the files on your computer (spreadsheets, documents, data)
- **Write** new files (apps, reports, charts)
- **Run** the things it writes — and show you what they do
- **Fix** what it gets wrong, when you point it out

You don't write code. You **describe what you want**, and you **check the result**.

What it cannot do (be honest)

- **Read your mind.** You still have to be specific.
- **Replace knowing what's worth building.** That's still your job.
- **Be trusted blindly.** You verify. Always.

This is a **tool**, not a replacement for thinking.

The demo: meet Maya Chen

The scenario

Maya Chen — Chief Innovation Officer at **MeridianCare Health Network**
4 hospitals · 18 clinics · Pacific Northwest

Nine months ago Maya launched **Sentinel Health** — a **Remote Patient Monitoring** pilot:

- **250 patients** across 6 partner clinics
- Chronic conditions: **Diabetes · Hypertension · CHF · COPD**
- Bluetooth devices · symptom app · weekly virtual check-ins
- **~\$1,800 per patient per year**

Maya's problem

The board meets in **4 weeks**. They will decide:

- **Scale system-wide** (~12,000 eligible patients), or
- **Sunset the program.**

She needs to answer five questions. Today.

The five questions

1. **Outcomes** — Are ER visits and hospitalizations actually down vs. the year before?
2. **Segmentation** — Which patient groups benefit most?
3. **Engagement** — Is engagement strong enough? Does it correlate with outcomes?
4. **Sites** — Which of the 6 clinics are working? Which aren't?
5. **Economics** — Does \$1,800/patient/year pay for itself?

The dashboard we'll build answers all five.

Here's the data she has

A spreadsheet. **250 rows. 26 columns.**

- Patient demographics (age, condition, baseline risk)
- Devices and engagement (readings, app logins, check-ins)
- Clinical metrics (A1C, blood pressure, weight, SpO2)
- Utilization (ER visits, hospitalizations — before & during)
- Cost (monthly, total to date)
- Patient satisfaction (1–5)

(Every patient in this dataset is **synthetic** — Elise generated it. No real patient data.)

Now we switch to Claude Code

What you're about to see: **me building this dashboard, live, from scratch.**

- No slides for the next ~25 minutes
- I'll narrate as I go
- You'll see the **real terminal**, the **real prompts**, the **real edits**, and the **dashboard appearing piece by piece**
- It will not be perfect on the first try. That is the point.

When something looks unfamiliar, stop me. The vocabulary on the next slide should cover most of it.

Vocabulary you'll see (1 of 2)

- **LLM** — *Large Language Model*. The AI behind the curtain. Reads text, writes text, makes decisions.
- **Claude** — the LLM I use. Made by Anthropic. (ChatGPT is OpenAI's; Gemini is Google's.)
- **Claude Code** — a tool that lets Claude read, write, and run files **on my laptop**, not just chat in a browser.
- **Terminal** — the black text window I type into. It's just another way to talk to your computer.

Vocabulary you'll see (2 of 2)

- **Git repo** — a folder on my laptop that **tracks every change**, so I can undo anything.
- **GitHub** — a website where Git repos live online; lets you share + back up your code.
- **Streamlit** — the tool that turns Python data code into a web dashboard with almost no extra work.

Don't worry about memorizing these — they'll come up in context.

Questions?

Contact + further reading

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The deck and the dashboard you saw are in a repo I'll share after the talk.

Merci à **Dr. Elise Lambert** et **Dr. Julia Fleck** pour l'invitation. Thank you all for your attention.

Appendix

(For the curious — not part of the talk)

Bonus: from app to portable knowledge

The dashboard answers **today's question** — Maya's question — about Sentinel Health.

A package format called **PAX** (*Portable Analytical eXpertise*) goes further: it wraps the *domain itself* — the concepts, the findings, the data, and the analysis playbook — so the next health system can run the same analysis on their own data and get a comparable answer.

Ask afterward if curious.

Bonus: how this is actually built

- **Claude Code** — the AI coding tool I used
- **Streamlit** — simplest way to make an interactive Python dashboard
- **Plotly** — the charts
- **Pandas** — the data layer
- **Marp** — these slides
- **Git** — version control; every change is undoable

You don't need to learn any of these to *use* what AI builds for you. Those are the names to google later.

Bonus: what's in the shared repo

- `data/` — the synthetic Sentinel Health CSV
- `slides/` — this deck (Marp markdown)
- `docs/` — five guides:
 - i. How to analyze the data with Claude Code
 - ii. How to build the dashboard (the prompt I used)
 - iii. How to update the dashboard
 - iv. Git & GitHub — the 10-minute version
 - v. What Claude Code actually is

No dashboard code is shipped. You generate it yourself by pasting the prompt in **guide #2**.